

Remote Communication via Huggable Interfaces - Behavior Synchronization and Social Presence

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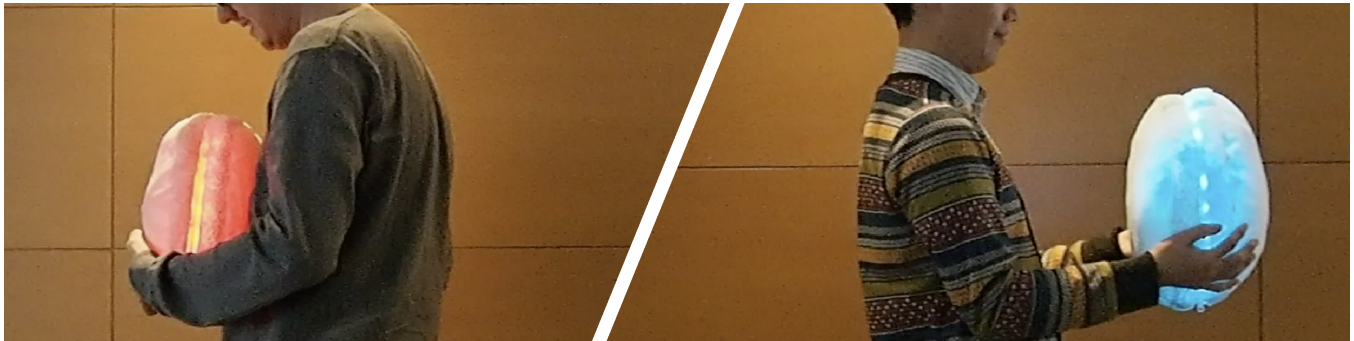


Figure 1: HugBits: a huggable interface to mediate remote affective communication. The interface senses the user's hug and conveys it to the partner using visual cues.

ABSTRACT

Interfaces for mediated communication are being designed to increase the quality of the interaction of people geographically separated. Communication modalities offered by the medium was found to affect on the perceived social presence, but also communication strategies involving higher coordinated actions could be related to a higher social presence. The purpose of this study was to investigate how behavior synchronization can influence individual's perceived social presence and behavior coordination while interacting through a huggable interface with simple visual cues. Through an experimental study that involved two virtual agents, it was observed that a partner that promote synchronous actions : 1) led to a higher level of perceived social presence, and 2) encouraged more responsive behaviors from the user.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices; Interaction design theory, concepts and paradigms.**

KEYWORDS

social presence, affective computing

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1 INTRODUCTION

New types of communication technologies are emerging, offering unique ways to interact and connect with others. Researchers have been looking at conventional communication media from different perspectives to propose solutions that increase the quality of communication among people who live geographically separated [5]. One approach argues that different media has different properties and abilities to produce rich social information [2], which commonly brings researchers to explore whether the perceived social presence levels change while interacting through one medium

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compared to others [12]. Among the different definitions found in the literature [1], social presence can be regarded as the degree of feeling connected to others through a medium, creating a positive, social and warm interaction [6]. For this reason, many studies focus on how different media modalities affect social presence [12], and how the representation of richer social cues (e.g., touch) impact on social presence.

A different approach refers to the individuals' ability of adapting to a given medium to achieve the intended communication. Individual's characteristics as well as communication strategies can have a significant effect on social presence [14]. For a medium that offers limited information, what kind of communication strategies lead to an enhanced feeling of social presence? One possible answer relies in the time relationship of the actions, based on the assumption that strategies that involve higher level of coordinated actions could be related to a higher social presence. For instance, interpersonal synchronization was found to influence the perceived social connection to others in online and face-to-face interaction [7]. Moreover, in mediated communication it was observed that media able to support synchronous communication creates higher levels of co-presence and satisfaction [9]. Measuring and understanding the quality of mediated interaction remains a challenge. It is necessary to investigate the way the user experiences the interaction through the interface, and the way the medium enables the sense of connectedness or closeness.

The main goal of this project is to develop a remote communication interface that enhances the affective communication and creates a positive experience. To achieve this, the current study revolves around the feeling of perceived social presence and its relationship with synchronized behaviors. Based on the different benefits that come with an enhanced feeling of social presence [6], interfaces for mediated communication are being designed to offer affordances that increase the levels of social presence. Following this approach, in the past [10] proposed a huggable interface to mediate remote communication. It offers a physical embodiment able to sense users' hugs to trigger messages represented as simple visual cues using colors. A follow up study compared a huggable interface with a screen-based medium [11]. The huggable interface was found more effective in creating a feeling of connectedness and positive affects, and it was inferred the possibility of this effect being related not only to the modality of the medium, but also to the synchronized actions between users. Can users perceive different experiences when they coordinate actions through a huggable interface with minimal information as well?

This sets the motivation of this study, for which we designed two virtual agents based on conditional probability model to manipulate the experiment conditions, allowing to compare users' experience with a "more interactive partner" and a "less interactive partner". We introduce a huggable interface for remote communication called "HugBits", designed to offer similar affordances to the ones found in the literature [10]. The purpose of this study is to investigate how behavior synchronization can influence individual's perceived social presence and behavior coordination while interacting through HugBits.

2 METHODOLOGY

2.1 HugBits

HugBits supports one-on-one remote communication using a set of paired cushions. It is designed for users that live apart and rely on media to establish communication, and provides an interface with physical embodiment that senses users' haptic information. Messages through HugBits are triggered by hugs (i.e., the user hugging the device), transferred to an identical device, and conveyed to the partner as visual cues made with colored light patterns. The nature of the messages sent with HugBits can be described by following the attributes introduced by [5]: 1) Messages open to interpretation without specific content, 2) Messages that can be seen by other people in proximity, 3) Ephemeral messages that disappear with time, and 4) Messages that contain information of synchronous and asynchronous actions.

2.1.1 System overview. HugBits consists of a soft and spherical cushion with the circuitry encapsulated in the center (Fig.1). HugBits includes three functionalities: sense haptic gestures, convey messages using visual cues, and support remote communication. The sensors are made of an array of 6 infrared photo reflective sensors that measure the deformation of the cushion. Sensing and hug detection were executed at 20Hz, and all sensor data was recorded by a mobile application. The information is then conveyed by an LED strip around the cushion. This information can indicate asynchronous or synchronous messages. Asynchronous messages refers to actions from both users that happen on different timings. In this case, HugBits slowly blinks in red when a hug is detected, and blue when a message is received. Synchronous messages are actions happening at the same time, and HugBits represents them as purple lights circulating around the cushion. Remote communication between two HugBits modules is managed by a mobile application and a server computer. The cushions pair via BLE with a mobile app designed to mediate the communication with the server via TCP/IP communication.

2.1.2 Virtual agent. Two virtual agents (*Sync/Async*) were implemented for this study, and they were designed to either promote or avoid synchronized hugs. Figure 2 illustrates the stochastic state transition models of the agents' behavior consisting of two states (*Hug/Not Hug*), and empirically selected transition probabilities according to the user's state. *Sync* agent promotes synchronized hugs by having a higher probability to change into *Hug* state when the user is hugging, and maintaining the synchronized hug once it's achieved. On the other hand, *Async* agent avoids synchronized hugs by having a lower probability to change into a *Hug* state while the user is hugging, and tends to terminate synchronized hugs when they occur. The transition probabilities when the user is not hugging are identical for both agents. During the experiment described in a later section, the agent's state transition was set to occur in a random interval between 2 and 5 seconds.

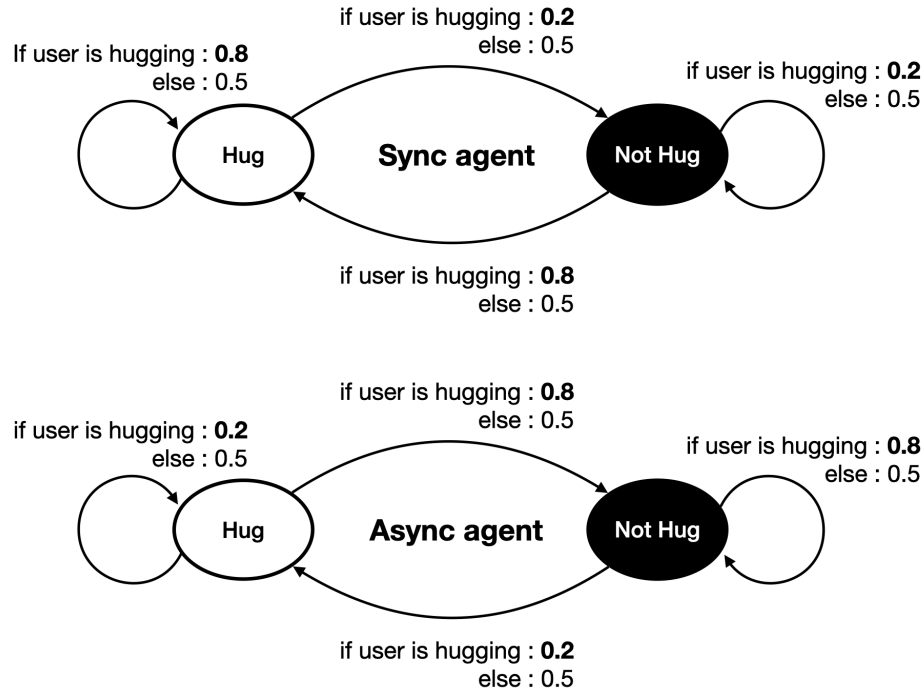


Figure 2: Stochastic state transition models of the virtual agent's behavior. The Sync agent (Top) tends to achieve/maintain the synchronized hug with the human user, whereas the Async agent (Bottom) tends to avoid the synchronized hug.

3 EXPERIMENT

3.1 Overview

To evaluate the effect of behavior synchronization on perceived social presence, we applied the two different agents described in Section 2.1.2 into HugBits. The following hypotheses will be addressed in this study:

Hypothesis 1 (H1): Communication with an agent that promotes synchronization contributes to higher perceived social presence.

Hypothesis 2 (H2): Individuals aim to coordinate their actions with the remote partner more when the agent is promoting synchronization.

3.2 Participants

Twenty participants (12 males, average age 21.05 ± 2.33) were recruited for participation. All participants were Japanese and college students, and they were recruited by a flyer shared in an internal mail list. The ethical committee of the authors' affiliation reviewed and approved this experiment. Written informed consent was obtained from all the participants in this study.

3.3 Experiment procedure

To prevent the spread of COVID-19 infections, participants were asked to wear a mask during the sessions, and to wash and sanitize their hands as soon as they arrived to the experiment room. They were asked to wear plastic gowns to minimize direct contact with the HugBits cushion.

The experiment was conducted by two experimenters for one participant at a time. One directly interacted with the participant (experimenter A), and the other was situated outside of the room (experimenter B). When the participant arrived to the experiment room, they were welcomed by both experimenters after which experimenter B left the room. Then, a video was shown to the participant explaining that HugBits can connect to people by detecting hugs and transferring them to an identical device remotely. In addition, an explanation on the meaning of colors displayed on a HugBits device was given.

For this experiment it was important to make the participants believe they were communicating with a human partner, and special effort was made to achieve this. The two experimenters A and B communicated remotely by voice message throughout the experiment, pretending that they were dealing with another participant simultaneously. Participants were explained that the purpose of the experiment was to evaluate their experience when they communicate with someone else using HugBits. They were asked to communicate with two different partners located in a different room in the building, and to fill questionnaires to evaluate the quality of their experience. In reality, the two partners were the two virtual agents implemented as a computer program (Section 2.1.2), and the questionnaires were evaluating perceived social presence (Section 3.4.1). Considering that the communication strategy and coordination are factors that directly influence on perceived social presence, a pre-programmed agent was chosen over a human partner to offer similar experience to every participant.

Each experiment consisted of two sessions, and each participant interacted with each virtual agent once. The order of the conditions was counterbalanced between the participants to cancel out the order effect. Each condition lasted 3 minutes, and participants were allowed to freely communicate using HugBits (i.e., by hugging the device). After each condition, the participants filled a questionnaire and after finishing the second condition they were asked about their experience in a short interview. Their interactions with the agents was recorded both by HugBits sensor data, and by a video camera.

3.4 Measurements

3.4.1 Questionnaire. Different social presence scales can be found in the literature, and some of them are more appropriated than others for certain interfaces. For instance, some scales aim to study social presence in virtual environments [4] or in human-agent interaction [13]. For a communication interface that provides minimal information, such as HugBits, we measured social presence as a combination of subjective scales that measure perceived other's co-presence [8], and affective attraction [3]. Both co-presence and affective attraction have been found to be closely related to social presence [4] [1] [12]. We found that the items describing perceived co-presence were more suitable to the experiment task. The scale consisted of 12 items rated using a five-point Likert scale. However, we excluded two items because they were not suitable for the task ("My interaction partner was unwilling to share personal information with me", and "My interaction partner made our conversation seem intimate"), as well as one duplicated item. The remaining 9 items were as follows:

- Q1 My interaction partner was intensely involved in our interaction.
- Q2 My interaction partner seemed to find our interaction stimulating.
- Q3 My interaction partner communicated coldness rather than warmth.
- Q4 My interaction partner created a sense of distance between us.
- Q5 My interaction partner seemed detached during our interaction.
- Q6 My interaction partner created a sense closeness between us.
- Q7 My interaction partner acted bored by our interaction.
- Q8 My interaction partner was interested in communicating with me.
- Q9 My interaction partner showed enthusiasm while communicating with me.

Affective attraction was measured by asking participants to rate how they felt about their partners using 13-point semantic-differential scales anchored by Unpleasant–Pleasant (Q1), Cold–Warm (Q2), Negative–Positive (Q3), Unfriendly–Friendly (Q4), and Distant–Close (Q5). [3]. Both questionnaires were translated into Japanese language.

3.4.2 Hugging behavior. The analysis of hugging behavior involves selecting features that describe the interaction of each pair. Firstly, we counted the occurrence and duration of participant's hugs and agent's hugs, and the number of synchronized hugs. Synchronized

hugs were counted from the moment both interfaces' sensors detected a hug to the moment the sensors on either interface deactivated. Synchronized hugs were also classified into two categories: synchronized hugs when the user responds to the agent's hug (User-respond), or synchronized hugs when the agent responds to the user's hug (Agent-respond). Figure 3 illustrates the interaction between one of the participants and the two agents, where each message (hug) is represented as a shadowed area. The overlapping areas are synchronized hugs, and the two categories can be distinguished based on who initiated the hug and who is responding to the hug. Based on this information, we estimated 2 features to describe the behavior of the participant during the interaction with an agent. As shown in the equations below, we defined as "Responsiveness" the ratio of the participant's hugs that were meant to be a response to the agent's hug, and "Sensitivity" as the ratio of the agent's hugs that were followed by the participant's hug. A higher Responsiveness means that the majority of the participant's hugs were an attempt to synchronize. Similarly, a higher Sensitivity infer that the participant showed a response to the majority of the agent's hugs.

$$Responsiveness = \frac{\# \text{ of user-respond sync hugs}}{\# \text{ of user hugs}}$$

$$Sensitivity = \frac{\# \text{ of user-respond sync hugs}}{\# \text{ of agent hugs}}$$

4 RESULTS

4.1 Social Presence Score

Figure 4 shows the average and standard deviation of the resulted social presence scores (right: *Copresence*, left: *Attraction*). One-way repeated measures multivariate analysis of variance (MANOVA) was applied to investigate the statistical differences for both questionnaires, where the behavioral condition of the virtual agent (hereinafter refereed as "condition") was regarded as the independent variable, and responses to the items in the questionnaires were regarded as dependent variables. As shown in figure 4, significant differences between the conditions were confirmed for both *Copresence* ($Wilks'\Lambda = 0.29, F(9, 12) = 3.34, p < 0.05, \eta^2 = 0.72$) and *Attraction* ($Wilks'\Lambda = 0.46, F(5, 16) = 3.81, p < 0.05, \eta^2 = 0.54$). In addition, post-hoc univariate analyses showed that there were significant differences in all questionnaire items between the two conditions.

4.2 Hugging Behavior Analysis

Figure 5 summarizes the participants' behavioral characteristics as basic quantities (i.e., total duration and occurrence of hugging behavior) and behavioral features defined in Section 3.4.2. The average and standard deviation of total duration and occurrence of hugging behavior are shown on the left, and those of *Responsiveness/Sensitivity* features are shown on the right. According to the result of a one-way repeated measures MANOVA, although there were no significant differences in duration and occurrence, there was a significant effect of the condition on behavioral features ($Wilks'\Lambda = 0.66, F(2, 18) = 4.65, p < 0.05, \eta^2 = 0.34$). Moreover, it was observed that both *Responsiveness* and *Sensitivity* increased significantly in the *Sync* condition.

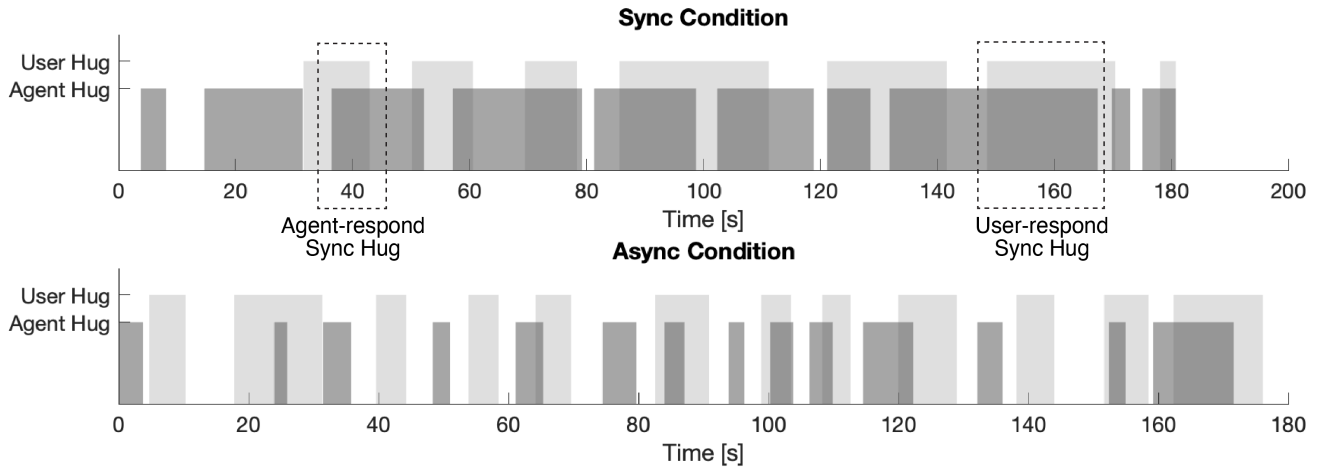


Figure 3: Example of hug behaviors in Sync condition (Top) and Async condition (Bottom) extracted from HugBits sensor data. Synchronized hugs were categorized into two types, Agent-respond sync hug and User-respond sync hug, according to who achieved the synchronized hug.

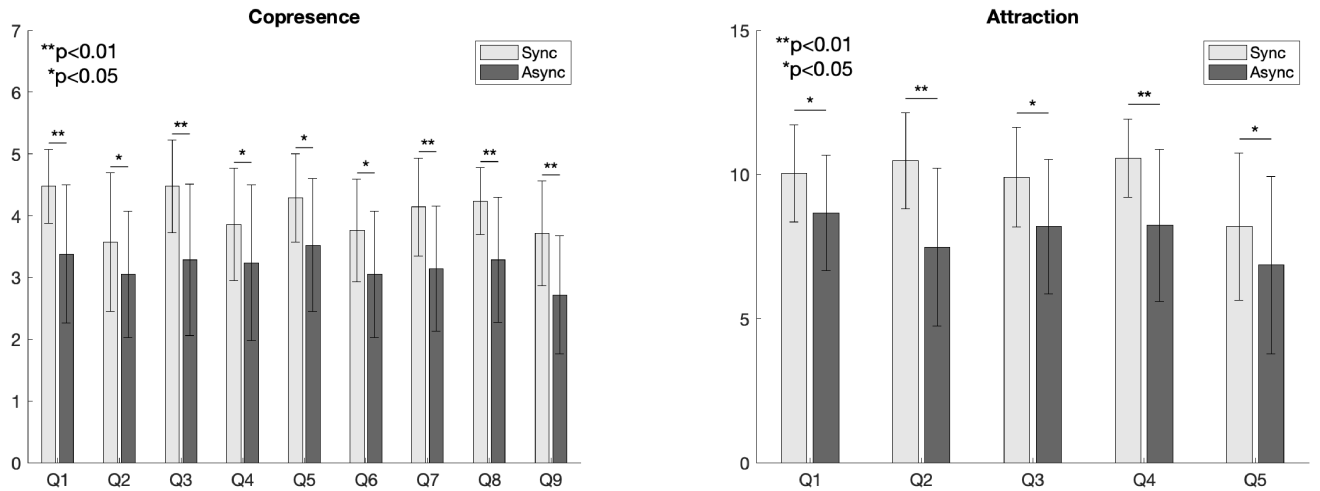


Figure 4: Comparison of social presence questionnaires between Sync and Async conditions : Co-presence(Left), Attraction (Right).

5 DISCUSSION

To test hypothesis 1 (H1) we relied on subjective report using two questionnaires (Section 3.4.1). The concepts of co-presence and affective attraction are closely linked to social presence, and they were found more suitable to evaluate HugBits in the experimental context. Results on Figure 4 shows a clear difference between the two agents, as significant difference was found on each of the items, supporting H1. The feeling of being in synchrony offered by very limited cues with no specific content, was enough to enhance the feeling of being connected, and resulted in a more positive experience. The second hypothesis (H2) refers to the effect on the user's behavior, and the analysis involves extracting features that describe the interaction. The number of synchronized hugs alone did not

give us much information about the user, since the agents were going to affect this value. By using features such as the described "Responsiveness" and "Sensitivity" it was possible to observe how much of the user's behavior was "social" or responsive to the partner's actions. This effect became more evident if we observe both plots presented on Figure 5. While the number and duration of users' hugs were higher on the asynchronous condition, they were less coordinated with the agent as responsiveness and sensitivity scored significantly lower (supporting H2). Follow up studies could consider investigating the use of different features that give a deeper understanding of the interaction. For example, information about the time to respond a message, or group synchronized actions with different duration could give another perspective that

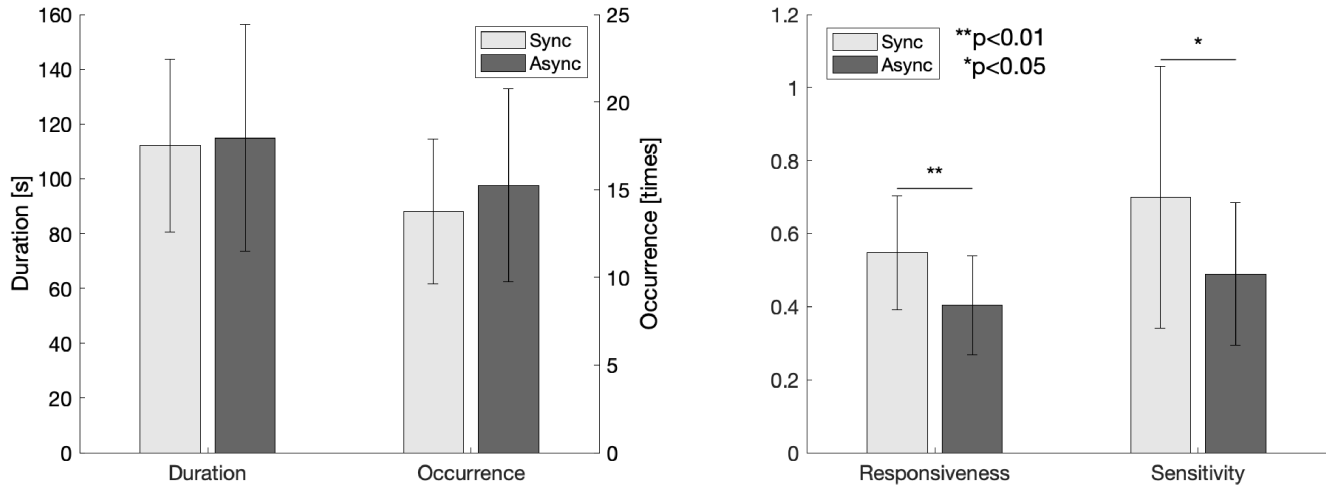


Figure 5: Comparison of the participants' behavioral features between *Sync* and *Async* conditions.

helps to understand the quality of the interaction. Another possible direction could handle observable events that contributes to predict subjective experiences such as social presence. Instead of relying solely on questionnaires, it might be beneficial to explain the effect of the interface based on the explanatory concept inferred from the observations.

Through the interviews, privacy concerns were raised as the communication through HugBits was considered to be rather intimate. To address this, one suggestion could be to support different haptic gestures to trigger the message. For example, a user could send a message by either hugging or shaking the cushion. This allows the user can choose how to communicate according to their context. Another suggestion is to rely on multimodal information to represent hugs. For instance, a soft vibration that can only be felt by the user, and colors that can be shared with others allow to keep an intimate channel and keep the visibility to create awareness and modify the physical space. Regarding the design of the agents, we observed that while the selected probabilities helped to create the desired experience, one participant among the group rated higher the asynchronous agent on both scales. During the interview, the participant explained that the partner (async agent) seemed to be a more playful person, and that they could improvise a kind of turn-taking game. The other partner (sync agent) was perceived more predictable and detached from the interaction. While these impressions were different from the general consensus, it raises interesting questions about the effect of individual aspects (e.g., personality, preferences, background) on the perception of the interface.

The need for communication systems designed to enhance the affective experience is there, and it is more relevant than ever. Regarding real world applications, we envision systems like HugBits not only to enhance conventional communication media (i.e., video, voice chat), but especially tailored for socially dislocated individuals. We found a clear example on hospitalized patients who in many cases have impediments to communicate or meet their families. A communication medium that creates a strong affective response

could be beneficial for them, making it necessary to understand its underlying mechanisms of mediated communication, and exploit them to maximize their effect.

6 CONCLUSION

Achieving social presence in remote communication settings is challenging due to the limited cues in traditional communication media. In this paper we introduced HugBits, a huggable interface to foster affective communication between geographically separated users, and investigated how synchronized hugging behavior affects perceptions of social presence and behavior coordination. Our results showed that participants perceived a higher social presence when they interacted with the agent that promoted synchronization. Moreover, it was observed that participants behavior tended to coordinate more with the agent that promoted synchronization. This report ended by providing future directions, design implications and potential applications for remote communication interfaces similar in purpose and functions to HugBits.

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